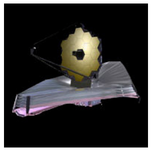

12063 - The Birth of a Cluster: A NIRSpec Survey of an X-ray Detected Proto-Cluster at z~5.7

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	Obs Wgt 100 max 7	NIRSpec MultiObject Spectroscopy	(1) JADES-ID1

ABSTRACT

JADES-ID1 is the highest-redshift (z ~ 5.7) X-ray-emitting proto-cluster known, seen merely ~1 Gyr after the Big Bang. Deep JWST imaging reveals a highly significant galaxy overdensity (66 candidate members) with a halo mass of log M ~ 13.3 Msun. Ultra-deep Chandra imaging reveals spatially extended soft X-ray emission co-spatial with the JWST galaxy overdensity, consistent with shock-heated intracluster medium (ICM). To transform this exciting detection into a physical portrait of early cluster assembly, we propose a NIRSpec MOS program to obtain systemic redshifts and rest-frame optical diagnostics for the full JADES-ID1 candidate list plus filler targets. In this program, we will (1) deliver a secure membership census and revise halo mass estimates, (2) map line-of-sight kinematics to identify subgroups and constrain dynamical assembly, (3) quantify environmental quenching by measuring H alpha (and H beta where available) as a function of radius and velocity offset, and (4) trace chemical enrichment and AGN incidence from rest-frame optical line ratios and broad-line diagnostics. The proposed observations will connect galaxy evolution to the formation of the ICM at the cosmic dawn and provide a solid reference for models of the earliest phases of cluster formation.

OBSERVING DESCRIPTION

We propose to measure the spectroscopic properties of the member galaxies of JADES-ID1, the highest redshift proto-cluster with X-ray emission known to date. These measurements will determine the precise redshift for each galaxy, and quantify the Halpha, [OIII] and possibly Hbeta, [NII] and other emission lines in the bandpass.

The best instrument to efficiently characterize the spectral properties of galaxies at redshift z~5.7 is NIRSpec Multi-Object Spectroscopy with its micro-shutter assembly (MSA).
We select our primary targets from the JADES-ID1 member list (66 galaxies in Li et al. 2024), and exclude the 9 galaxies with existing spectroscopic measurements, and the 3 galaxies that were approved for NIRSpec MOS observations in cycle 4. This results in 54 primary targets. Our filler targets are based on the Rieke et al. (2023) photometric catalog, and have to be within the 1.5arcmin of the centroid of the member galaxy distribution, and should not have existing spectroscopic measurements in the JWST or VLT/MUSE archive, which adds 871 filler targets. All our targets have precise positions from prior JWST NIRCam imaging.

Using the MSA Planning Tool we find an optimal distribution for the MSA setup with 7 exposures, with 2 shutter slitlet (nod in slitlet). We choose the NRSIRS2RAPID readout pattern and MSATA target acquisition. Smart scheduling predicts 7 visits, all of which produce a lower threshold violation for the data excess. However, we do not require pure parallels, making this a tolerated violation.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	JADES-ID1	RA: 03 32 33.5624 (53.1398433d) Dec: -27 47 43.43 (-27.79540d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 12063, Observation 1: Obs Wgt 100 max 7

Fri Mar 13 22:06:19 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Obs Wgt 100 max 7 (Obs 1)) Warning (Form): Config c2 (#2) has 1 filler slits affected by failed closed shutters.
(Obs Wgt 100 max 7 (Obs 1)) Warning (Form): Config c6 (#6) has 1 filler slits affected by failed closed shutters.
(Visit 1:1) Warning (Form): Data Excess over lower threshold
(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Data Excess over lower threshold
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Data Excess over lower threshold
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Data Excess over lower threshold
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Data Excess over lower threshold
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:6) Warning (Form): Data Excess over lower threshold
(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:7) Warning (Form): Data Excess over lower threshold
(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	JADES-ID1	RA: 03 32 33.5624 (53.1398433d) Dec: -27 47 43.43 (-27.79540d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
4		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
5		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
6		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
7		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	Primary (54 sources)	Filler (871 sources)	jwst-nirspec-g395m	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395M/F290LP)	c1	2 Shutter Slitlet	53.13835654166667 Degrees -27.80361194444447 Degrees	100.00068091970144			2	8	10970.845
2	1 (G395M/F290LP)	c2	2 Shutter Slitlet	53.14988533333333 Degrees -27.787358333333316 Degrees	99.99534278029041			2	8	10970.845
3	1 (G395M/F290LP)	c3	2 Shutter Slitlet	53.13377025 Degrees -27.788229166666667 Degrees	100.00283220176918			2	8	10970.845
4	1 (G395M/F290LP)	c4	2 Shutter Slitlet	53.15919720833333 Degrees -27.78542611111112 Degrees	99.9910166818604			2	8	10970.845
5	1 (G395M/F290LP)	c5	2 Shutter Slitlet	53.13897333333333 Degrees -27.794118055555543 Degrees	100.00040607999088			2	8	10970.845
6	1 (G395M/F290LP)	c6	2 Shutter Slitlet	53.14101491666666 Degrees -27.815873055555585 Degrees	99.99943037492227			2	8	10970.845
7	1 (G395M/F290LP)	c7	2 Shutter Slitlet	53.126821666666665 Degrees -27.781503888888892 Degrees	100.00607002479761			2	8	10970.845

Special Requirements

Group Visits within 53.0 Days
Visits Same PA
MSA Planned Aperture PA 100.0000 to 100.0000 Degrees (V3 321.4254303 to 321.4254303)